

GAMS assignment

Design and solve an optimization model using GAMS that reflects a real-world or interest-based decision problem of your choice. This project allows you to apply modeling skills to a context that is engaging to you — whether it relates to your everyday life, a personal hobby, a professional activity, or topics covered in class.

Requirements:

1. Propose a decision-making problem that can be formulated as an optimization model (e.g., linear, or mixed-integer-linear programming).
2. Clearly define:
 - sets, parameters, and decision variables,
 - an objective function (e.g., cost minimization, profit maximization),
 - constraints based on the logic of the problem.
3. Use real, estimated, or randomly generated data to populate your model.
4. Implement the model in GAMS and solve it.
5. Analyze and interpret the results:
 - Are the results meaningful?
 - How sensitive are the results to changes in key assumptions?
6. Prepare a short written report (maximum 3 pages) including:
 - description of the problem and its motivation,
 - mathematical model,
 - input data,
 - GAMS code (may be placed in the appendix),
 - results and interpretation,
 - conclusions and suggestions for improvement.